

Template Engine: The User Journey

F9 Info Technologies

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Template Engine: The User Journey

Project: Subhadra Group website (`cmt15bmts000401s6fn5ydhzp`) Audience: the admin who builds a client website with Studio (the Template Engine). Items that could not be verified in a browser are marked **unverified**.

1. The journey at a glance

5 Sign in	Admin panel opens on the Dashboard
1 Select Platform	Sidebar > Application Settings opens "Step 1 – Select Platform"
Platform"	Pick ONE platform per project. Web Application = the active platform.
2 Company details	Application Settings > Brand Profile: company name, emails, phones, addresses
3 Modules check	Sidebar > Modules: Template Engine, Theme Engine, Page Builder,
Builder,	Menus, Sectors, Catalog, Work, FAQ are switched on
4 Template Engine	Sidebar > Template Engine (opens the default project)
5 Overview	Upload logo, confirm company and contact details [Intake]
6 Color Palette	Check the 4 brand colors taken from the logo [Palette]
7 Inference and	AI suggests fonts and tone, a person approves the kit [Inference, Approval]
Approval	Brand colors become design tokens in the Theme Engine
8 Brands > Web app	Typography > Font settings > Navigation > Pages [Website]
Layout	Choose Top header, Header and Footer designs
10 Fill the content	Sectors, Catalog, Work, FAQ, Case Studies, Team
page	Each Sector, Catalog and Work entry gets its own detail
11 Menus	Fine-tune header and footer menus
12 Edit pages	Page editor and Section Builder
13 Finish	Guidelines PDF, Collateral, Preflight, Export [optional]
14 Go live and	Open the public pages, and re-run only what changed
maintain	

2. Before you start

You need	Where to check
An admin account with these permissions: <code>template-engine:view</code> and <code>:run</code> ; plus <code>:approve</code> for Approval and <code>:export</code> for Export	Sidebar > Access Control > Roles / Permissions
Modules switched on: Template Engine, Theme Engine, Page Builder, Menus, Sectors, Catalog, Work, FAQ, Case Studies	Sidebar > Modules (a module that is off has no sidebar entry)
A project to build in	The Studio opens the default project automatically. There is no sidebar screen for creating projects (unverified how new ones are created).
Object storage for uploaded files	Sidebar > Application Settings > Storage

3. Step by step

Each step shows where you click, what you see and do, what the system does behind the scenes, and what you get.

Start. Sign in

- **You:** log in to the admin panel.
- **You see:** the Dashboard. The left sidebar lists Dashboard, Users, Modules, then every switched-on module, then the groups Media, Access Control and Application Settings.
- **Next:** go to Application Settings.

Step 1. Select Platform (first screen after you click Application Settings)

After signing in, the user clicks **Application Settings** (the Theme Engine / Studio entry). The first screen is **Step 1 — Select Platform**. It asks one question: **what are you building?** The rule on the screen is *one platform per project*. Global design tokens stay shared across all platforms, and each platform exposes only the settings that belong to its own ecosystem.

- **You see:** platform cards grouped by family:

Group	Guideline	Platforms
Web	Responsive Web + WCAG	Website, Landing Page, Web Application , Admin Dashboard, Progressive Web App (PWA)
Mobile	Material Design 3 / Apple HIG	Android Native, iOS Native
Tablet	MD3 / Apple HIG, large screens	Android Tablet, iPad (iPadOS)
Desktop	Multi-window and keyboard navigation	Windows Desktop, macOS Desktop, Linux Desktop
TV	10-foot UI, remote / D-Pad focus	Android TV, Google TV, Apple TV (tvOS), Samsung Smart TV (Tizen), LG Smart TV (webOS), Roku TV, Amazon Fire TV
Wearables	Compact glanceable layouts	Wear OS, watchOS (Apple Watch), Samsung Galaxy Watch

- **You do:** click **Web Application**. This is the **active platform** for the journey in this document.
- **What happens next:** the flow continues exactly as described in the following steps (company details, modules, Overview, Color Palette, Approval, Brands > Web app, Layout, content, menus, editor). The choice of Web Application is what makes the Brands screen show the **Web app** card, and it is the platform whose design tokens the Approval stage writes to the Theme Engine.
- **Other platforms:** they are shown so the client sees the full roadmap, but they are **not active**. Picking one does not start a flow yet.
- **Where this stands in the current build:** the 27-platform picker is **not in the code today** (I searched the whole repository). What exists now is: (a) the Studio opens straight on Overview with Web App already fixed, (b) the Brands screen shows a single **Web app** card, and (c) the Theme Engine page has a platform bar with Web App, TV, Android Native App and iOS Native App. The picker in the screenshot is the intended first step, still to be built.

Step 2. Application Settings (company details)

- **Click:** sidebar > **Application Settings**. The group opens with four items:
 - **Types:** named groups of settings (for example Brand Profile, Theme Settings).
 - **Categories:** sub-groups inside a type.
 - **Fields:** the individual settings (text boxes, images, colors) that belong to a type.
 - **Storage:** where uploaded files are kept.
- Every active **Type** also appears as its own sidebar item. Open **Brand Profile** (`/admin/settings/view/brand-profile`).
- **You do:** enter the company name (required), primary and secondary email, primary and secondary phone, and address lines 1 and 2.
- **Behind the scenes:** the values are saved as setting fields of the `brand-profile` type. They are seeded by `backend/prisma/seeder/brand-profile-fields.seed.js` and are **global**, not per project.
- **Why here first:** the Studio's Overview screen shows these same fields. Filling them here means the Overview opens pre-filled, and the header, footer and Contact page pick them up. You can also fill them later, directly in the Overview screen.
- **Result:** company details are stored once and reused everywhere.

Step 3. Confirm the modules

- **Click:** sidebar > **Modules**.
- **You check:** Template Engine, Theme Engine, Page Builder (Section Builder), Menus, Sectors, Catalog, Work, FAQ and Case Studies are on.
- **Why:** the Studio depends on Theme Engine, Page Builder, Brand Kit, Collateral and Credits. Sectors, Catalog and Work are independent modules that plug in through the detail-page registry.

Step 4. Open the Template Engine

- **Click:** sidebar > **Template Engine** (`/admin/template-engine`).
- **What happens:** the page looks up your default project and sends you to `/admin/template-engine/projects/<projectId>/intake`.
- **First time:** if the project has no run yet, you see **Start Studio run**. Click it (creates the run, needs `template-engine:run`).
- **This project:** run `cmt18teqh000101rxwfzfn-dow` already exists and is **COMPLETED**. Its id becomes the prefix of every generated page address (`/p/te-cmt18teqh000101rxwfzfn-dow-home`).

- **You see:** a top bar with three tabs: **Overview**, **Color Palette**, **Brands**. Six more stages exist but have no tab (see step 7).

Step 5. Overview (Intake)

`/admin/template-engine/projects/<projectId>/intake`

- **You see:** one card, **Logo & Contact Details**.
- **You do:**
 1. Click **Upload logo** (svg, png, jpeg or webp). A preview and a status badge (draft, extracted, inferred, approved) appear.
 2. Check or enter the company name and contact fields (the same fields from step 2).
 3. Click **Next**.
- **Behind the scenes:**
 - The logo is cleaned, virus-scan queued, stored as media, and also becomes the admin panel logo (sidebar header).
 - **Next** checks that company name and logo exist, saves the fields, marks the INTAKE stage done, and opens Color Palette.
 - If the Website stage was already built, saving here quietly re-runs it so the logo and contact details reach the existing pages.
- **Result:** a brand kit exists for the project, with logo and contact data.

Step 6. Color Palette

`.../palette`

- **What happens on its own:** the palette is extracted from the logo (color math, no AI).
- **You see:** four colors: Primary, Secondary, Tertiary (accent) and Quaternary (neutral), with generated shade ramps.
- **You do:** keep them, or change each by typing a hex code, clicking **Randomize**, or using the eyedropper to pick from the logo. Click **Submit palette**.
- **Behind the scenes:** the palette is saved to the brand kit. A kit that is already approved cannot be changed.
- **Result:** brand colors are fixed. The screen moves on to **Brands** (Website).

Step 7. Inference and Approval (hidden tabs)

These two stages have no tab. They are reached by their address, or were already done.

Stage	Address	What the person does	Behind the scenes
Inference	<code>.../inference</code>	Click Run stage ; read the suggested typography pair and tone	Calls the AI service (falls back to a rule table if unavailable); charges 10 credits
Approval	<code>.../approval</code>	Tick each contrast adjustment, click Approve (needs <code>template-engine:approve</code>)	Marks the brand kit approved; writes the colors and type settings into the Theme Engine as design tokens and locks them

- **Result:** the whole web app now uses the client's brand tokens. You can see them under sidebar > **Theme Engine**.
- **Important:** the server will not build the Website until Inference and Approval are done. On this project they are done. On a new project, the Palette screen's jump to Brands skips them, so run them first (**unverified in a browser**).

Step 8. Brands > Web app (Website stage)

`.../website`. Only one brand card exists: **Web app**. It has four steps.

8A. TYPOGRAPHY

- **You do:** pick heading fonts (Poppins, Inter, Manrope, Space Grotesk) and body fonts (Inter, Roboto, Open Sans, Work Sans), or add your own font files from the Media picker. Click **Next**.
- **Saved:** to the brand kit.

8B. FONT SETTINGS

- **You do:** set sizes for H1 to H6, body, navigation and buttons.
- **Saved:** in your browser only (local storage).

8C. NAVIGATION

- **You do:**
 - Tick **suggested pages** (Home, About, Products and more), or type a **custom page**, or add an **external link**.
 - Drag to reorder and nest items into drop-down menus. Use the globe toggle to copy an item into the footer menu.
 - Click a **detail-page chip** (Sector detail, Work detail, Catalog detail) to create the parent menu item (“Sectors”, “Work”, “Products & Services”) with one child per entry.
- **Behind the scenes:** these are the real **Header** and **Footer** menus, the same rows you see in sidebar > Menus. A custom page label creates a menu item that points to `/p/te-<runId>--<page-name>`.
- **Rule:** names that match a detail-page type or a sector, work or catalog entry are not created as ordinary pages, because those get their own detail pages.

8D. PAGES (ASSEMBLE)

- **What happens on entering the step:** the Website stage runs with your navigation and layout choices.
- **The system:**
 - creates each selected page (default: Home, About, Contact) from the built-in template pack (general, medical or construction; the screen does not offer a choice, so the default applies),
 - fills in your brand colors, logo, contact details and layout,
 - publishes each new page.
- **Existing pages** are never rebuilt. Only navigation links, brand data, contact details, logos and layout are refreshed.
- **You see:** a grid of page cards, each with **Edit**. Also **Demo all pages** (opens the home page) and a **Layout settings** panel.
- **Result:** live pages at `/p/te-<runId>--<page>`.

Step 9. Layout

- **Click:** in the Layout settings panel, **Create layout** or **Edit layout** (`.../website/layout`).
- **You see:** three cards: **Top header**, **Header**, **Footer**. Each has an on/off switch and four designs, shown as live tiles filled with the client’s own logo and details.
- **You do:** pick a design per card. Use **View** for a full-size preview, **Edit** to open the Home page in the editor, or **Create new** to design a new block in the Section Builder. Click **Create layout**.

- **Behind the scenes:** the choice is saved and the Website stage runs again, applying the layout to every page.
- **Back on the Pages step** you will also see pills such as **Layout Sector detail**, **Layout Work detail** and **Layout Catalog detail**. Each opens the shared detail page in the editor, and its number badge opens the list of entries.

Step 10. Fill in the content modules

Each has a sidebar item. Open it with the project selected (`?projectId=<id>` is added for you, or the default project is used).

Sidebar item	What you add	What the system creates	Public address
Sectors	Sector name, description, image, SEO	A detail page from the shared Sector template	<code>/sectors/<slug></code>
Catalog	Product or service name, category, description, image, brand tag	A detail page from the shared Catalog template	<code>/catalog/<slug></code>
Work	Work category (Central AC, Home Theater and so on)	Its own detail page	<code>/work/<slug></code>
FAQ	Questions and answers	Shown in the quote-form section of pages	(inside pages)
Case Studies	Project stories	Shown in the Projects slider on the Home page	(inside pages)
Team	People	Shown in team and leadership blocks	(inside pages)

- **List actions on Sectors, Catalog and Work: Edit page** opens the shared detail page in the editor (changing the layout there changes every entry), the **gear icon** edits that entry's own name, text, image and tag, and the **arrow icon** opens the public page.
- **Shared layout, own content:** Sector and Catalog entries share one page layout per project, but each keeps its own text and images.
- **Menus:** to list the entries in the header menu, go back to Navigation and click the detail-page chip (step 8c).

Step 11. Menus

- **Click:** sidebar > **Menus**, choose Header Navigation or Footer Navigation.
- **You do:** rename, reorder, hide or nest items.
- **Result:** the public header and footer read the live menu on every page load, so no stage needs to be re-run.

Step 12. Edit pages

- **Open: Edit** on a page card, or **Edit page** in the Sectors, Catalog or Work list. Address: `/admin/template-engine/edit/<pageId>?projectId=<id>`.
- **You see:** a visual editor with a Section list on the left and the page preview on the right. Header, top bar and footer are shown but locked; change them only through Layout (step 9).
- **You do:** add, reorder and style sections (Hero Slider, Welcome, Inner Banner, About, Counters, Services, Featured Projects, Sectors, Team and more), edit text and images, then publish.
- **Section Builder** (sidebar > Section Builder) is for designing new custom blocks.

Step 13. Finish the run (optional stages)

Stage	Address	What it does
Guidelines	<code>.../guidelines</code>	Renders a brand guidelines PDF (5 credits). Skippable.
Collateral	<code>.../collateral</code>	Renders a visiting card and letterhead (credits per render). Skippable.
Preflight	<code>.../preflight</code>	Checks that none of the three branches failed
Export	<code>.../export</code>	Produces the export manifest (needs <code>template-engine:export</code>)

- When every stage is done or skipped, the run becomes **COMPLETED** and the Studio releases its locks on the Theme Engine fields.

Step 14. Go live and maintain

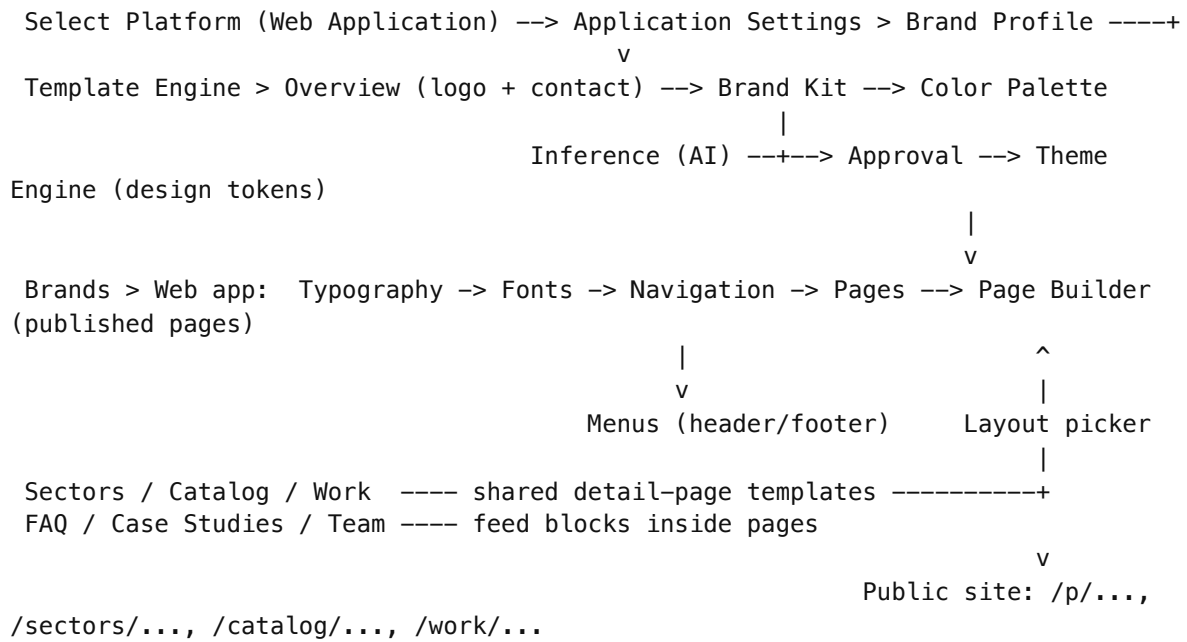
Public pages:

Address	What it shows
<code>/p/te-<runId>-home</code> , <code>-about</code> , <code>-contact</code> and so on	The assembled pages
<code>/sectors/<slug></code>	A sector detail page
<code>/catalog/<slug></code>	A product or service detail page
<code>/work/<slug></code>	A work detail page

What to re-run after a change:

You changed	Do this
Logo or contact details	Save in Overview (it re-runs Website automatically if Website is done)
Navigation pages	Brands > Web app > Navigation, then move to Pages to re-build links
Header, top bar or footer design	Layout > Create layout
Sector, Catalog or Work text and images	Edit the entry in its own list (no re-run needed)
Menu order or labels	Sidebar > Menus (no re-run needed)
Page sections	Edit in the page editor and publish
Brand colors or fonts	Color Palette or Typography, then Approval, then re-run Website

4. How the screens connect



5. Screen and address reference

Screen	Sidebar path	Address
Dashboard	Dashboard	<code>/admin/dashboard</code>
Modules	Modules	<code>/admin/modules</code>
Select Platform (Step 1)	Application Settings (first screen)	Intended screen, not in the current code. Today: Theme Engine platform bar at <code>/admin/theme-engine</code>
Application Settings	Application Settings > Types / Categories / Fields / Storage	<code>/admin/settings/types</code> , <code>/categories</code> , <code>/fields</code> , <code>/storage</code>
Brand Profile	Application Settings > (Type) Brand Profile	<code>/admin/settings/view/brand-profile</code>
Template Engine	Template Engine	<code>/admin/template-engine</code>
Overview (Intake)	(tab)	<code>/admin/template-engine/projects/<id>/intake</code>
Color Palette	(tab)	<code>.../palette</code>
Inference, Approval	(no tab)	<code>.../inference</code> , <code>.../approval</code>
Brands > Web app	(tab)	<code>.../website</code>
Layout picker	Pages step > Create layout	<code>.../website/layout</code>
Page editor	Pages step > Edit	<code>/admin/template-engine/edit/<pageId>?projectId=<id></code>
Section Builder	Section Builder	<code>/admin/page-builder/section-builder</code>
Theme Engine	Theme Engine	<code>/admin/theme-engine</code>
Menus	Menus	<code>/admin/menus</code>

Screen	Sidebar path	Address
Sectors, Catalog, Work	Sectors, Catalog, Work	<code>/admin/sectors</code> , <code>/admin/catalog</code> , <code>/admin/work</code>
FAQ, Case Studies, Team	FAQ, Case Studies, Team	<code>/admin/faq</code> , <code>/admin/projects-content</code> , <code>/ad- min/team</code>
Media library	Media > Library	<code>/admin/media</code>

6. What happens behind the scenes (summary)

The Studio is a 9-stage pipeline. Order is enforced by the server.

```

INTAKE -> PALETTE -> INFERENCE -> APPROVAL --> GUIDELINES (optional) --+
                                         +-> COLLATERAL (optional) ---+>
PREFLIGHT -> EXPORT
                                         +-> WEBSITE      (optional) --+
  
```

- The engine owns two tables only: runs and stages. Everything else lives in the modules it calls.
- Stages run inside the web request. There is no background queue, and no cancel button.
- Only three things cost credits: brand inference (10), guidelines PDF (5) and each collateral render.
- Every run route is scoped to the project. A run from another project is not visible.

Stage	Module it calls
Intake	Brand Kit
Palette	Brand Kit (color extraction)
Inference	Brand Kit > AI service; Credits
Approval	Brand Kit; Theme Engine
Guidelines	Brand Kit; Credits; storage
Collateral	Collateral; Credits
Website	Brand Kit; Media; Page Builder; detail-page registry (Sectors, Work, Catalog)
Preflight, Export	(read other stages)

7. Permissions along the journey

Step	Permission
Open Studio, read runs	<code>template-engine:view</code>
Start run, advance, retry, skip	<code>template-engine:run</code>
Approval stage	<code>template-engine:approve</code>
Export stage	<code>template-engine:export</code>
Application Settings	<code>types:view</code> , <code>categories:view</code> , <code>setting-fields:view</code> , <code>settings:view</code>
Section Builder / page editor	<code>page-builder:view</code>
Sectors, Catalog, Work	<code>sectors:*</code> , <code>catalog:*</code> , <code>work:*</code> (view, add, edit, delete)
Menus	<code>menus:*</code>

8. Things to watch for

1. **Palette jumps to Brands.** On a new project, run Inference and Approval first, or Website is refused with “stage gate failed”.
2. **Brand Profile is global.** Changing it here changes it for every project.
3. **Website only refreshes pages, never rebuilds them.** To get new built-in content on an existing page, edit it in the editor.
4. **Font settings live in the browser.** Clearing browser data resets them.
5. **Inference is generic.** It does not receive industry or company name, so its suggestions are not industry specific.
6. **Run status.** A failed stage does not mark the run failed, and there is no cancel.
7. **Body-copy AI** for Hero and About is designed but not built.
8. **Hidden tabs.** Inference, Approval, Guidelines, Collateral, Preflight and Export have no tab; use the addresses in step 7 and step 13.

9. Loading the client-approved content (for the team)

Approved Subhadra content is loaded with repeatable scripts in `backend/scripts` (run from `backend/` with `DATABASE_URL` set). Order:

1. Content: `seed-catalog-content.js <projectId>`, `seed-sector-content.js`, `seed-services-content.js`, `seed-about-content.js`, `seed-home-content.js`, `seed-products-services-content.js`, `seed-sectors-page-content.js`, `seed-leadership-content.js`, `seed-contact-content.js`, `seed-work-content.js`.
2. Site chrome: `seed-site-chrome.js <projectId>`.
3. Links: `fix-internal-links.js <projectId>` (last).

Data files are in `backend/scripts/seed-data/`; images in `frontend/public/seed/subhadra/`.

10. Main files

Area	Path
Engine	<code>backend/src/modules/template-engine/</code>
Studio screens	<code>frontend/src/app/admin/template-engine/</code> and <code>.../_components/stages/</code>
Sidebar	<code>frontend/src/components/layout/AdminSidebar.tsx</code>
Application Settings	<code>frontend/src/app/admin/settings/</code>
Detail-page registry and templates	<code>backend/src/shared/detail-pages/</code>
Design documents	<code>.agents/arch/TEMPLATE_ENGINE_ARCH.md</code> , <code>docs/superpowers/specs/</code>